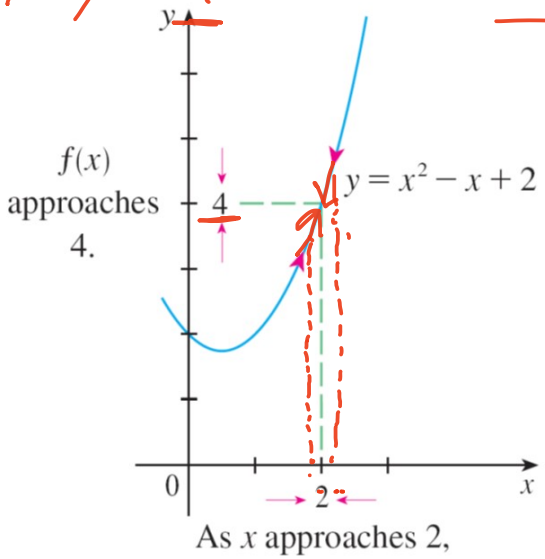
CHAPTER 2 LIMITS AND DERIVATIVES

2.2 The Limit of a Function

- Calculus is the study of limits.
- How do we find the limit of $x^2 - x + 2$ as x approaches 2? $\lim_{x \to 2} x^2 - x + 2 = 4$

$x \to 2^-$	$f(x)$	$x \to 2^+$	$f(x)$
1.0	2.000000	3.0	8.000000
1.5	2.750000	2.5	5.750000
1.8	3.440000	2.2	4.640000
1.9	3.710000	2.1	4.310000
1.95	3.852500	2.05	4.152500
1.99	3.970100	2.01	4.030100
1.995	3.985025	2.005	4.015025
1.999	3.997001	2.001	4.003001

$f(x) \to 4$ $\quad$ $f(x) \to 4$



As x approaches 2,

- **Definition**

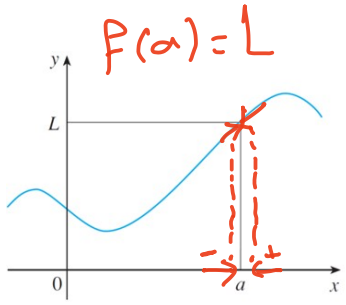
Suppose that $f(x)$ is defined when x is *near* the number a. We write

$$\lim_{x \to a} f(x) = L$$

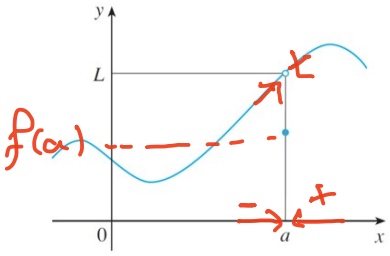
and say that

the **limit** of $f(x)$, as x approaches a, equals L

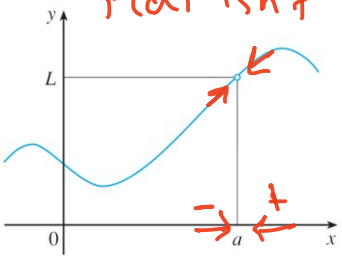
if we can make the values of $f(x)$ arbitrarily close to L by restricting x to be sufficiently close to a (on either side of a) but *not equal* to a.



(a)



(b)



(c)

Example 2.1. Find $\lim_{x \to 3} (4x - 5)$.

x	$4x - 5$
2.999	
3.001	

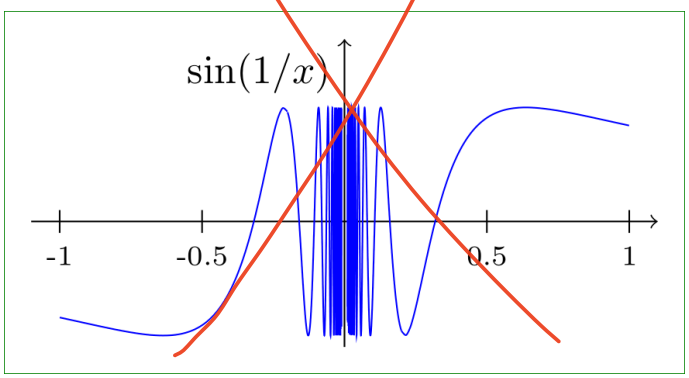
Example 2.2. Find $\lim_{x\to 3} \dfrac{x^2-x-6}{x-3}$.

x	$\dfrac{x^2-x-6}{x-3}$
2.999	
3.001	

Example 2.3. Find $\lim_{x\to 0} \dfrac{\sqrt{x^2+9}-3}{x^2}$.

x	$\dfrac{\sqrt{x^2+9}-3}{x^2}$
± 0.01	
± 0.001	
± 0.0001	
± 0.00001	

Example 2.4. Find $\lim_{x\to 0} \sin\left(\dfrac{1}{x}\right)$.



- **Definition**

We write

$$\lim_{x \to a^-} f(x) = L$$

and say that

the **left-hand** limit of $f(x)$, as x approaches a from the left, equals L if we can make the values of $f(x)$ arbitrarily close to L by restricting x to be sufficiently close to a with x less than a.

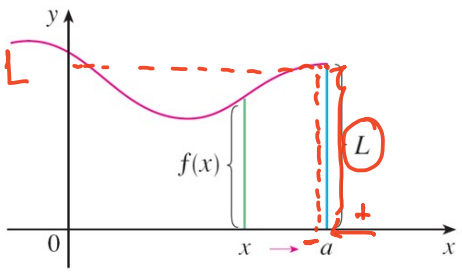
- **Definition**

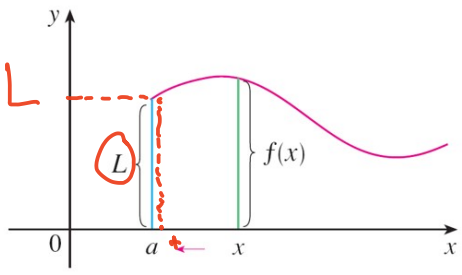
We write

$$\lim_{x \to a^+} f(x) = L$$

and say that

the **right-hand** limit of $f(x)$, as x approaches a from the right, equals L if we can make the values of $f(x)$ arbitrarily close to L by restricting x to be sufficiently close to a with x greater than a.

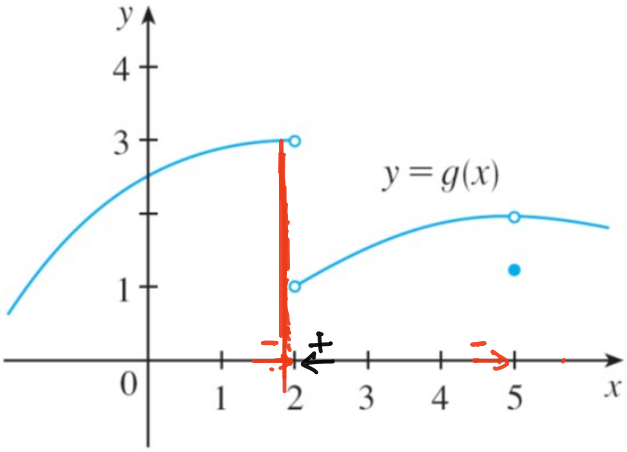




- **Theorem**

$\lim_{x \to a} f(x) = L$ if and only if $\lim_{x \to a^+} f(x) = L$ and $\lim_{x \to a^-} f(x) = L$.

Example 2.5. The graph of a function g is shown below. Use it to state the values (if they exist) of the following:



(a) $\lim_{x \to 2^-} g(x) = 3$

(b) $\lim_{x \to 2^+} g(x) = 1$

(c) $\lim_{x \to 2} g(x)$ Does not exist, since $\lim_{x \to 2^-} g(x) \neq \lim_{x \to 2^+} g(x)$

ONE

(d) $\lim_{x \to 5^-} g(x) = 2$

(e) $\lim_{x \to 5^+} g(x) = 2$

(f) $\lim_{x \to 5} g(x) = 2$

(g) $g(5) = 1$

- The Heaviside function H is define by

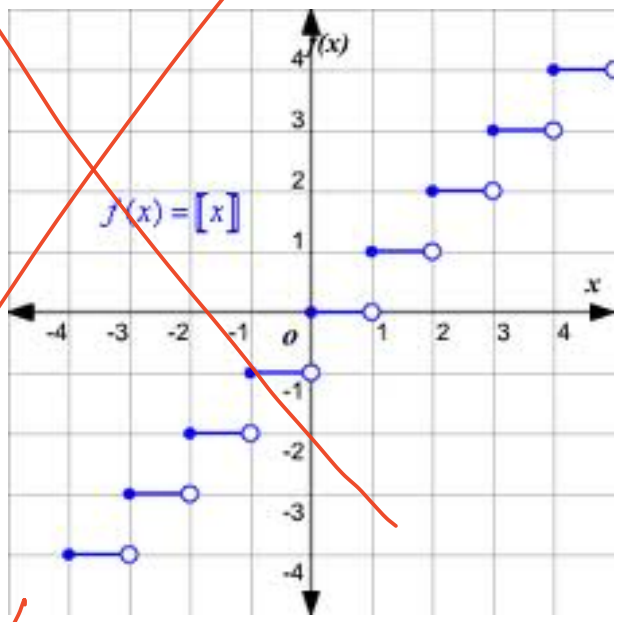
$$H(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$

- Find $\lim_{t \to 0^-} H(t)$ and $\lim_{t \to 0^+} H(t)$.

$\lim_{t \to 0^-} H(t) = 0$

$\lim_{t \to 0^+} H(t) = 1$

- $[\![x]\!]$ denotes the greatest integer less than or equal to x.



Example 2.6. Find $\lim_{x \to 2} [\![x]\!]$.

$\frac{L}{0} = \mp\infty$

Infinite Limits

Example 2.7. Find $\lim_{x\to 0} \frac{1}{x^2}$ if it exists.

$\lim_{x\to 0^+} \frac{1}{x^2} = \infty$

$\lim_{x\to 0^-} \frac{1}{x^2} = \infty$

$\therefore \lim_{x\to 0} \frac{1}{x^2} = \infty$

$\lim_{x\to 0} \frac{1}{x^2}$ DNE

- **Intuitive Definition of an Infinite Limit**

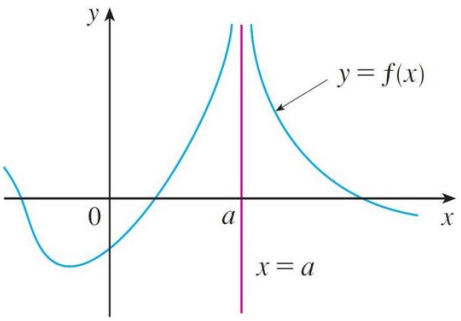
Let f be a function defined on both sides of a, except possibly at a itself. Then

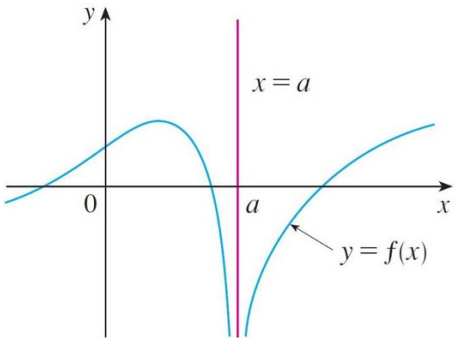
$$\lim_{x\to a} f(x) = \infty$$

means that the values of $f(x)$ can be made arbitrarily large by taking x sufficiently close to a, but not equal to a.

$$\lim_{x\to a} f(x) = -\infty$$

means that the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to a, but not equal to a.





$\lim_{x\to a^+} f(x) = \infty$

$\lim_{x\to a^-} f(x) = \infty$

$\lim_{x\to a} f(x) = \infty$

Example 2.8. Find $\lim_{x\to 3^+} \frac{1}{x-3}$ and $\lim_{x\to 3^-} \frac{1}{x-3}$ if they exist.

$\lim_{x\to 3^+} \frac{1}{x-3} = +\infty$

$x > 3$

What if $x = 3.00001$?

$\lim_{x\to 3^-} \frac{1}{x-3} = -\infty$

What if $x = 2.9999$?

$x < 3$

$\lim_{x\to 3} \frac{1}{x-3}$ D.N.E

Example 2.9. Find $\lim_{x\to -3^+} \frac{2x}{x+3}$ and $\lim_{x\to -3^-} \frac{2x}{x+3}$.

$\lim_{x\to -3^+} \frac{2x\ (-)}{x+3\ (+)} = -\infty$

$x > -3$

What if $x = -2.999$

$\lim_{x\to -3^-} \frac{2x\ (-)}{x+3\ (-)} = \infty$

$x < -3$

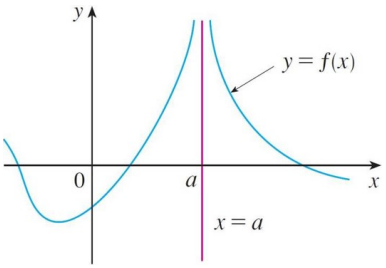
What if $x = -3.0001$

- **Vertical Asymptotes** V.A

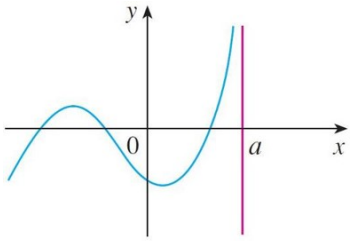
The vertical line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true.

1. $\lim_{x \to a} f(x) = \infty$ 2. $\lim_{x \to a^-} f(x) = \infty$ 3. $\lim_{x \to a^+} f(x) = \infty$

4. $\lim_{x \to a} f(x) = -\infty$ 5. $\lim_{x \to a^-} f(x) = -\infty$ 6. $\lim_{x \to a^+} f(x) = -\infty$

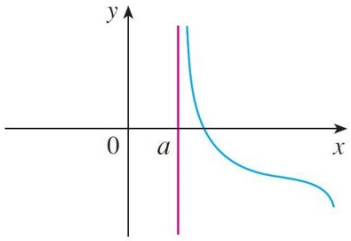


$\lim_{x \to a} f(x) = \infty$



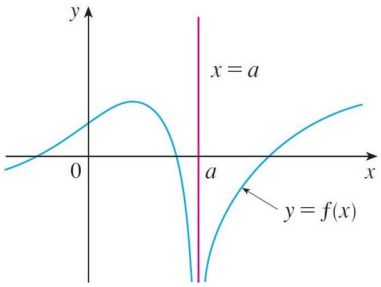
(a) $\lim_{x \to a^-} f(x) = \infty$

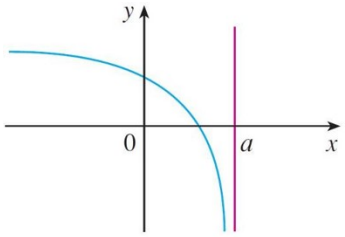
$x = a$ is V.A



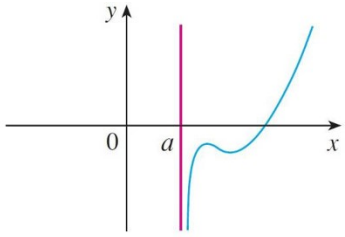
(b) $\lim_{x \to a^+} f(x) = \infty$

$x = a$ is V.A





(c) $\lim_{x \to a^-} f(x) = -\infty$

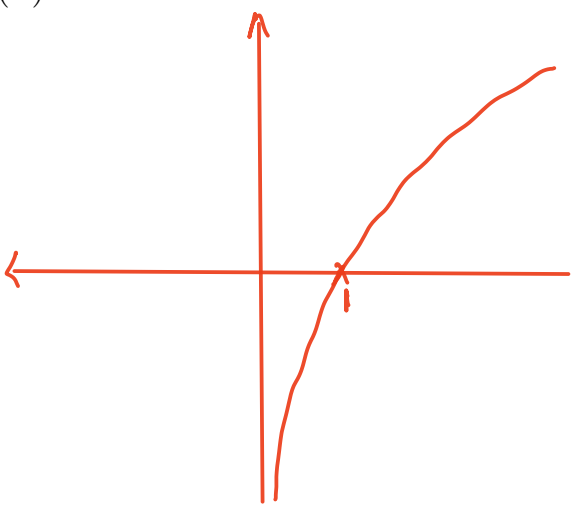


(d) $\lim_{x \to a^+} f(x) = -\infty$

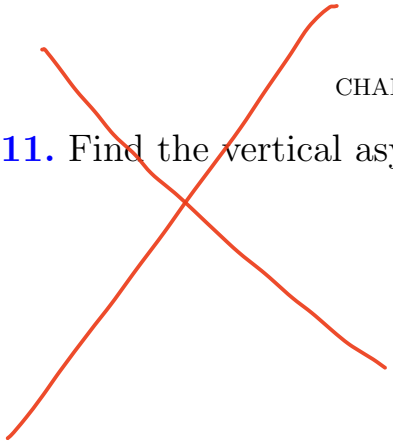
Example 2.10. Find the vertical asymptote of $f(x) = \ln x$.

Since, $\lim_{x \to 0^+} \ln x = -\infty$

$\therefore x = 0$ is V.A of $f(x)$



Example 2.11. Find the vertical asymptotes of $f(x) = \tan x$.



Exercises Page 92 5, 7, 9, 11, 19, 23, 27, 31, 33, 35, 41, 43, 45(a) + find the vertical asypmtote of the function.

2.3 Calculate Limit Using the Limit Laws

- **Limit Laws**

Let c be a constant, and f and g be functions that have limits at a. Then

1. $\lim_{x\to a} cf(x) = c\lim_{x\to a} f(x)$.
2. $\lim_{x\to a} [f(x) + g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$.
3. $\lim_{x\to a} [f(x) - g(x)] = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)$.
4. $\lim_{x\to a} [f(x) \cdot g(x)] = \lim_{x\to a} f(x) \cdot \lim_{x\to a} g(x)$.
5. $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$ provided that $\lim_{x\to a} g(x) \neq 0$.
6. $\lim_{x\to a} c = c$.
7. $\lim_{x\to a} x = a$.
8. $\lim_{x\to a} [f(x)]^n = \left[\lim_{x\to a} f(x)\right]^n$, n is a positive integer.
9. $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)}$.

Example 2.12. Find $\lim_{x\to 3} 2x^4$.

$$\lim_{x\to 3} 2x^4 = 2\left[\lim_{x\to 3} x\right]^4 = 2(3)^4 = 162$$

DIY

Example 2.13. Find $\lim_{x\to 4}\left(3x^2-2x\right)$. $= 40$

DIY

Example 2.14. Find $\lim_{x\to 4}\dfrac{\sqrt{x^2+9}}{x}$.

T

Example 2.15. If $\lim_{x\to 3} f(x)=4$ and $\lim_{x\to 3} g(x)=8$, find $\lim_{x\to 3}\left[f^2(x)\cdot\sqrt[3]{g(x)}\right]$.

$$\lim_{x\to 3}\left[f^2(x)\cdot\sqrt[3]{g(x)}\right]=\lim_{x\to 3}\left[f(x)\right]^2\cdot\lim_{x\to 3}\sqrt[3]{g(x)}$$

$$=\left[\lim_{x\to 3} f(x)\right]^2\sqrt[3]{\lim_{x\to 3} g(x)}=(4)^2\sqrt[3]{8}=16(2)=32$$

- **Substitution Property**

> If f is a **polynomial function** or a **rational function**, then
>
> $$\lim_{x\to a} f(x)=f(a)$$
>
> provided that $f(a)$ is defined.

DIY

Example 2.16. Find $\lim_{x\to 2} \dfrac{7x^5 - 10x^4 - 13x + 6}{3x^2 - 6x - 8}$.

Example 2.17. Find $\lim_{x\to 1} \dfrac{x^3 + 3x + 7}{x^2 - 2x + 1}$. $= \dfrac{11}{0}$

$\lim_{x\to 1^+} \dfrac{x^3+3x+7}{x^2-2x+1} = +\infty$, let $x = 1.0001$

$\lim_{x\to 1^-} \dfrac{x^3+3x+7}{x^2-2x+1} = \infty$, let $x = 0.999$

- **Theorem**

> Suppose $f(x) = g(x)$ for all x in an open interval containing the number a, except possibly at the number a itself, and $\lim_{x\to a} g(x)$ exists, then $\lim_{x\to a} f(x)$ exists and $\lim_{x\to a} f(x) = \lim_{x\to a} g(x)$.

$=$

DIY

Example 2.18. Find $\lim_{x\to 1}\dfrac{x^2-1}{x-1}$. $= 2$

Example 2.19. Find $\lim_{x\to 2}\dfrac{x^2+3x-10}{x^2+x-6}$. $= \dfrac{0}{0}$

$x+3$, $x-2$ | $x+5$, $x-2$

$$\lim_{x\to 2}\frac{x^2+3x-10}{x^2+x-6}=\lim_{x\to 2}\frac{(x+5)(x-2)}{(x+3)(x-2)}=\lim_{x\to 2}\frac{x+5}{x+3}=\frac{7}{5}$$

Example 2.20. Given $g(x)=\begin{cases}\dfrac{x-2}{x^2-4}, & x\neq 2\\ 6, & x=2\end{cases}$, find $\lim_{x\to 2} g(x)$.

$\frac{0}{0}$

$$\lim_{x\to 2} g(x)=\lim_{x\to 2}\frac{x-2}{x^2-4}=\lim_{x\to 2}\frac{x-2}{(x-2)(x+2)}=\lim_{x\to 2}\frac{1}{x+2}=\frac{1}{4}$$

original	conjugate
$a+b$	$a-b$
$a-b$	$a+b$

$x=\sqrt{x}\sqrt{x}$

Example 2.21. Find $\lim_{x\to 1}\dfrac{x-1}{\sqrt{x}-1}$. $= \dfrac{0}{0}$

Method ① :-

$$\lim_{x\to 1}\frac{x-1}{\sqrt{x}-1}=\lim_{x\to 1}\frac{(\sqrt{x}-1)(\sqrt{x}+1)}{\sqrt{x}-1}$$

$$=\lim_{x\to 1}\sqrt{x}+1=2$$

Method ② :-

$$\lim_{x\to 1}\frac{x-1}{\sqrt{x}-1}\times\frac{\sqrt{x}+1}{\sqrt{x}+1}$$

$$=\lim_{x\to 1}\frac{(x-1)(\sqrt{x}+1)}{x-1}$$

$$=\lim_{x\to 1}\sqrt{x}+1=2$$

DIY

Example 2.22. Evaluate $\lim_{x\to 0}\dfrac{\sqrt{x^2+9}-3}{x^2}$. $= \frac{1}{6}$

Ex: ① Absolute Function ② Squre root function

- Some limits are best calculated by first finding the **left-** and **right-**hand limits.

Example 2.23. Find the limit $\lim_{x\to 0}|x|$.

$$\lim_{x\to 0^+}|x| = \lim_{x\to 0^+} x = 0$$

$$\lim_{x\to 0^-}|x| = \lim_{x\to 0^-} -x = 0$$

$$\therefore \lim_{x\to 0}|x| = 0$$

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Example 2.24. Does the limit $\lim_{x\to 0}\dfrac{|x|}{x}$ exists?

$$\lim_{x\to 0^+}\frac{|x|}{x} = \lim_{x\to 0^+} 1 = 1$$

$$\lim_{x\to 0^-}\frac{|x|}{x} = \lim_{x\to 0^-} -1 = -1$$

$$\therefore \lim_{x\to 0}\frac{|x|}{x} \text{ D.N.E}$$

$$\frac{|x|}{x} = \begin{cases} \frac{x}{x} = 1, & x > 0 \\ \frac{-x}{x} = -1, & x < 0 \end{cases}$$

DIY

Example 2.25. Given $f(x) = \begin{cases} \sqrt{x-2}, & x > 6 \\ x-4, & x < 6 \end{cases}$, find $\lim_{x\to 6} f(x)$. $= 2$

Example 2.26. Find $\lim_{x \to -2} [\![x]\!]$.

Suppose $f(x) \leq g(x)$ for all x in an open interval containing the number a, except possibly at the number a itself, then $\lim_{x \to a} f(x) \leq \lim_{x \to a} g(x)$.

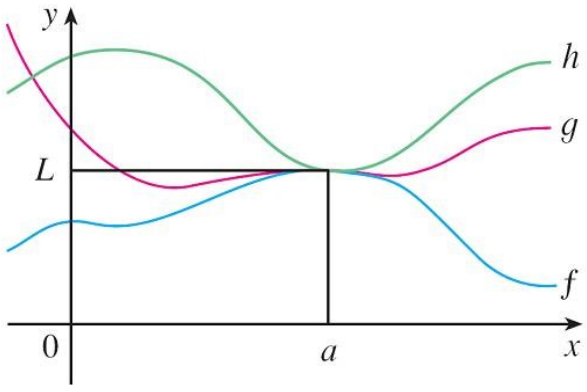
- **Squeeze Theorem/Sandwich Theorem**

Suppose $f(x) \leq g(x) \leq h(x)$ for all x in an open interval containing the number a, except possibly at the number a itself, and

$$\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L$$

then

$$\lim_{x \to a} g(x) = L$$



DIY

Example 2.27. Find $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right)$. = 0

$-1 \le \sin\theta \le 1$

Example 2.28. Find $\lim_{x\to 0} x \sin\left(\frac{1}{x}\right)$.

$-1 \le \sin\frac{1}{x} \le 1$ *

① When $x < 0$, $x \to 0^-$

$\Leftrightarrow\Leftrightarrow$

$-x \ge x\sin\frac{1}{x} \ge x$

$x \le x\sin\frac{1}{x} \le -x$

$\lim_{x\to 0^-} x = 0$ & $\lim_{x\to 0^-} -x = 0$

By squeeze theorem

$\therefore \lim_{x\to 0^-} x\sin\frac{1}{x} = 0$

② When $x > 0$, $x \to 0^+$

$\Leftrightarrow\Leftrightarrow$

$-x \le x\sin\frac{1}{x} \le x$

$\lim_{x\to 0^+} -x = 0$ & $\lim_{x\to 0^+} x = 0$

=

By squeeze theorem

$\lim_{x\to 0^+} x\sin\frac{1}{x} = 0$

$\therefore \lim_{x\to 0} x\sin\frac{1}{x} = 0$

Exercises Page 102 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25, 27, 29, 31, 37, 39, 41, 43, 47, 51, 53, 59, 61 (*Hint*: $0 \le f(x) \le x^2$ for all x.).

2.5 Continuity of Functions

- **Definition**

We say that f is continuous at a number a if

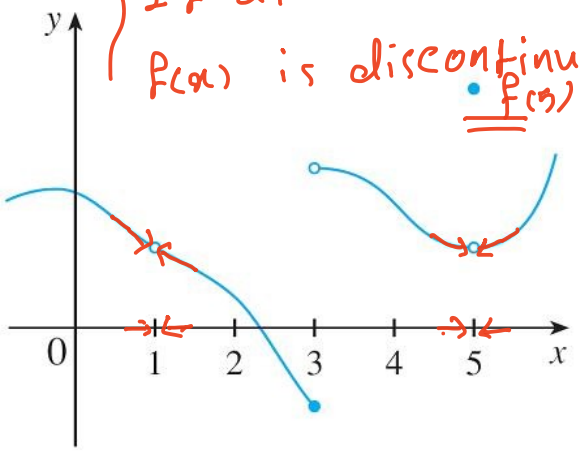
$$\lim_{x \to a} f(x) = f(a)$$

① $\lim_{x \to a} f(x)$ exist

② $f(a)$ is defined

③ $\lim_{x \to a} f(x) = f(a)$

If at least one condition fails, the $f(x)$ is discontinuous at a



At $x = 1$, $\lim_{x \to 1} f(x)$ exist but $f(1)$ D.N.E

At $x = 3$, $\lim_{x \to 3} f(x)$ D.N.E but $f(3)$ exist

At $x = 5$, $\lim_{x \to 5} f(x)$ and $f(5)$ exist but $\lim_{x \to 5} f(x) \neq f(5)$

$\therefore$ f is continuous $\forall x \in \mathbb{R} \setminus \{1, 3, 5\}$

DIY

Example 2.29. Determine whether the function $f(x) = \dfrac{1}{x-2}$ is continuous at $x = 2$.

Example 2.30. Determine whether the function $f(x) = \dfrac{x^2-4}{x-2}$ is continuous at $x = 2$.

$D_{f(x)} = \mathbb{R} \setminus \{2\}$

$\therefore$ $f(x)$ is not continuous at $x=2$ since $f(x)$ is undefined

Example 2.31. Determine whether the function $f(x) = \begin{cases} \dfrac{x^2-4}{x-2}, & x \neq 2 \\ 2, & x = 2 \end{cases}$ is continuous at $x = 2$.

① $\lim\limits_{x\to 2} f(x) = \lim\limits_{x\to 2} \dfrac{x^2-4}{x-2} = \lim\limits_{x\to 2} \dfrac{(x-2)(x+2)}{x-2}$ (0/0)

$= \lim\limits_{x\to 2} x+2 = 4$ $\neq$

② $f(2) = 2$

$\therefore$ $f(x)$ is discontinuous at $x=2$ since $\lim\limits_{x\to 2} f(x) \neq f(2)$

Example 2.32. Let $f(x) = \begin{cases} \dfrac{x^2-4}{x-2}, & x \neq 2 \\ c, & x = 2 \end{cases}$. Find the value of c such that f is continuous at $x = 2$ (*removing the discontinuity*).

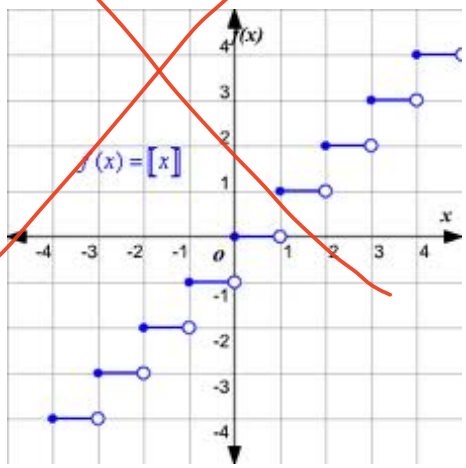
In order to let $f(x)$ continuous at $x = 2$,

$\lim_{x \to 2} f(x) = f(2)$

$4 = c$

Example 2.33. Determine all points of discontinuity of $f(x) = \dfrac{\sin x}{x(1-x)}$.

Example 2.34. Determine all points of discontinuity of $f(x) = [\![x]\!]$.



- **Definition**

We say that f is **left continuous** at a number a if

$$\lim_{x \to a^-} f(x) = f(a)$$

We say that f is **right continuous** at a number a if

$$\lim_{x \to a^+} f(x) = f(a)$$

- **Continuity on an Interval**

We say that f is continuous on an open interval if it is continuous at each point of that interval. It is continuous on the closed interval $[a, b]$ if it is continuous on (a, b), right continuous at a and left continuous at b.

Example 2.35. Show that the function $f(x) = 1 - \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$.

① Check on $(-1, 1)$

$D_{f(x)} = [-1, 1]$

$\therefore$ $f(x)$ is continuous on $(-1, 1)$

② Check on right continuous at $x = -1$

$$\lim_{x \to -1^+} f(x) = \lim_{x \to -1^+} 1 - \sqrt{1 - x^2} = 1 = f(-1)$$

$\therefore$ $f(x)$ is right continuous at $x = -1$

③ Check on left continuous at $x = 1$

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} 1 - \sqrt{1 - x^2} = 1 = f(1)$$

$\therefore$ $f(x)$ is left continuous at $x = 1$

$\therefore$ $f(x)$ is continuous on $[-1, 1]$

- **Theorem**

If f and g are continuous at a, and k is a constant, then the following functions are also continuous at a:

- kf
- $f + g$
- $f - g$
- $f \cdot g$
- $\dfrac{f}{g}$, provided that $g(a) \neq 0$
- f^n
- $\sqrt[n]{f}$, provided that $f(a) > 0$.

- **Examples of Continuous Functions**

7 Theorem The following types of functions are continuous at every number in their domains:

- polynomials
- rational functions
- root functions
- trigonometric functions
- inverse trigonometric functions
- exponential functions
- logarithmic functions

Example 2.36. Use continuity to evaluate the limit.

(a) $\lim_{x \to \pi} \sin x = f(\pi) = \sin \pi = 0$

(b) $\lim_{x \to 4} 3^x = f(4) = 3^4 = 81$

- **Composite Limit Theorem**

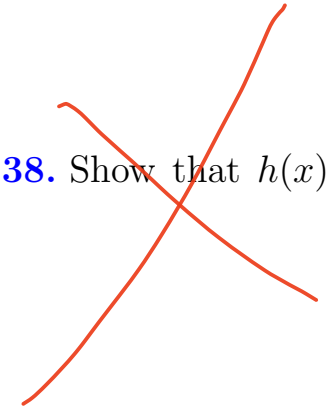
If $\lim_{x\to a} g(x) = L$ and f is continuous at L, then

$$\lim_{x\to a} f\left(g(x)\right) = f\left(\lim_{x\to a} g(x)\right) = f(L)$$

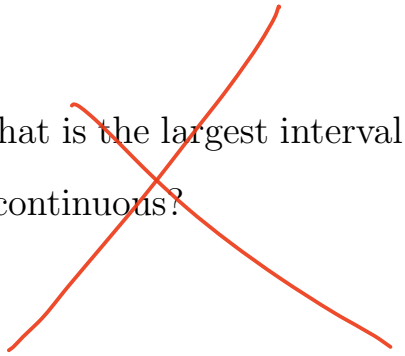
In particular, if g is continuous at a and f is continuous at $g(a)$, then the composite $f \circ g$ is continuous at a.

Example 2.37. Show that $h(x) = \left|x^2 - 3x - 6\right|$ is continuous at each real number.

Example 2.38. Show that $h(x) = \sin \dfrac{x^4 - 3x + 1}{x^2 - x - 6}$ is continuous except at 3 and -2.

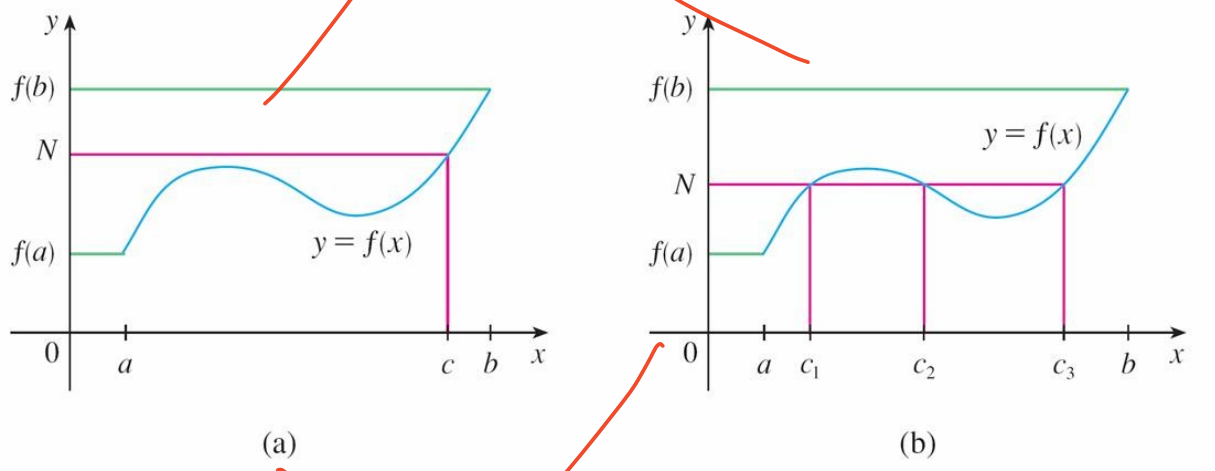


Example 2.39. What is the largest interval over which the function defined by $g(x) = \sqrt{4 - x^2}$ is continuous?



- **Intermediate Value Theorem**

> Let f be a function defined on $[a, b]$ and let N be a number between $f(a)$ and $f(b)$. If f is continuous on $[a, b]$, then there is at least a number c between a and b such that $f(c) = N$.



Example 2.40. Show that the equation $x - \cos x = 0$ has a solution between $x = 0$ and $x = \dfrac{\pi}{2}$.

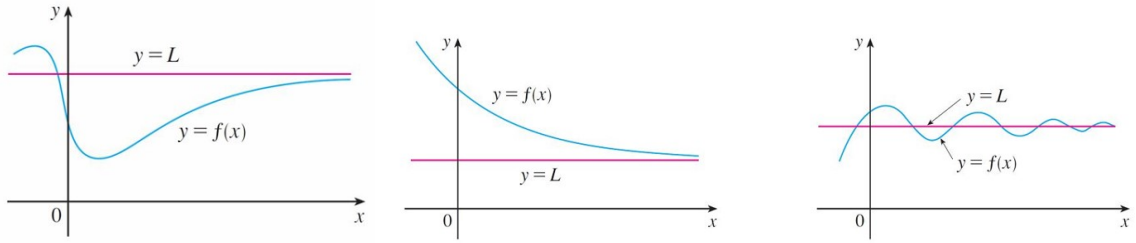
Exercises Page 124 3, 11, 13, 15, 17, 19, 21, 23, 27, 35, 37, 39, 41, 45, 47, 51, 53, 55, 69 (*Hint*: $x^3 + 1 = x$).

2.6 Limits at Infinity, Horizontal Asymptotes

- **Definition**

$$\lim_{x\to\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large.



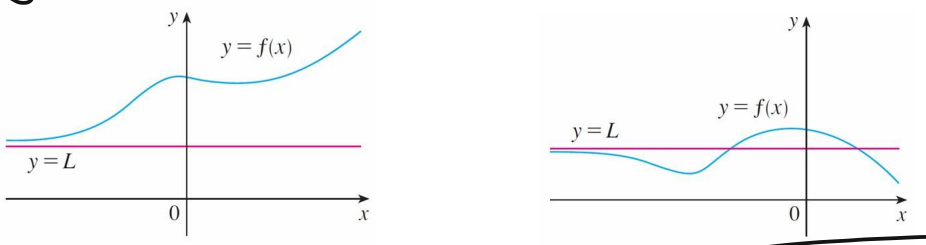
$$\lim_{x\to-\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large negative.

$\infty + \infty = \infty$

$\infty \cdot \infty = \infty$

$\frac{L}{\mp\infty} = 0$



Example 2.41. Find $\lim_{x\to\infty} \frac{x}{1+x^2}$. $= \frac{\infty}{\infty}$

indeterminate form $\frac{\mp\infty}{\mp\infty}, \frac{0}{0}, \infty\cdot 0, \infty-\infty$

$\lim_{x\to\infty} \frac{x}{x^2\left(\frac{1}{x^2}+1\right)} = \lim_{x\to\infty} \frac{1}{x\left(\frac{1}{x^2}+1\right)} = 0$

DIY

Example 2.42. Find $\lim_{x\to\infty} \dfrac{2x^3}{1+x^3}$. $= 2$

Example 2.43. Find $\lim_{n\to\infty} \sqrt{\dfrac{n+1}{n+2}}$. $= \dfrac{\infty}{\infty}$

$$\lim_{n\to\infty} \sqrt{\frac{n+1}{n+2}} = \lim_{n\to\infty} \sqrt{\frac{n\left(1+\frac{1}{n}\right)}{n\left(1+\frac{2}{n}\right)}} = \lim_{n\to\infty} \sqrt{\frac{1+\frac{1}{n}}{1+\frac{2}{n}}} = 1$$

$$(x-a)(x+a) = x^2 - a^2$$

Example 2.44. Evaluate $\lim_{x\to\infty} \left(\sqrt{x^2+2x} - x\right)$. $= \infty - \infty$

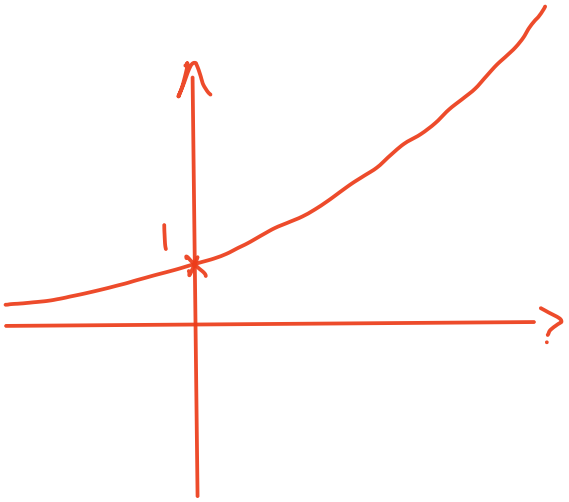
$$\lim_{x\to\infty} \sqrt{x^2+2x} - x \times \frac{\sqrt{x^2+2x}+x}{\sqrt{x^2+2x}+x} = \lim_{x\to\infty} \frac{x^2+2x-x^2}{\sqrt{x^2+2x}+x}$$

$$= \lim_{x\to\infty} \frac{2x}{\sqrt{x^2+2x}+x} \quad \left(\frac{\infty}{\infty}\right) = \lim_{x\to\infty} \frac{2x}{\sqrt{x^2\left(1+\frac{2}{x}\right)}+x} = \lim_{x\to\infty} \frac{2x}{|x|\sqrt{1+\frac{2}{x}}+x}$$

$$= \lim_{x\to\infty} \frac{2x}{x\sqrt{1+\frac{2}{x}}+x} = \lim_{x\to\infty} \frac{2x}{x\left[\sqrt{1+\frac{2}{x}}+1\right]} = \lim_{x\to\infty} \frac{2}{\sqrt{1+\frac{2}{x}}+1}$$

$$= \frac{2}{2} = 1$$

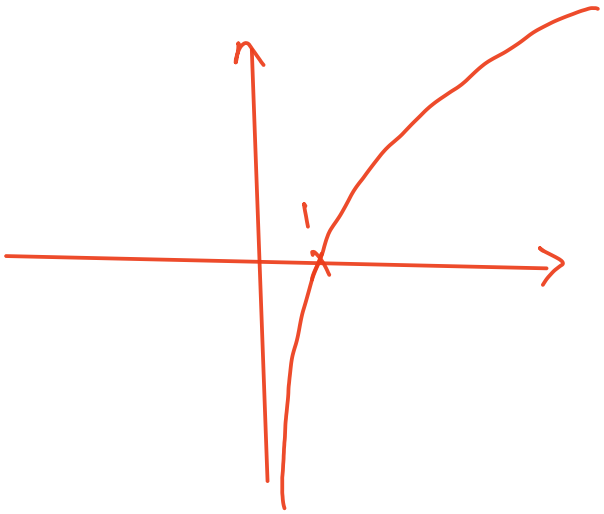
Example 2.45. Evaluate $\lim_{x\to 2^+} \arctan\left(\frac{1}{x-2}\right)$.



Example 2.46. Find $\lim_{x\to\infty} e^x$ and $\lim_{x\to-\infty} e^x$.

$\lim_{x\to\infty} e^x = \infty$

$\lim_{x\to-\infty} e^x = e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0$



Example 2.47. Find $\lim_{x\to\infty} \ln x$ and $\lim_{x\to 0^+} \ln x$.

$\lim_{x\to\infty} \ln x = \infty$

$\lim_{x\to 0^+} \ln x = -\infty$

Example 2.48. Find $\lim_{x\to 0^-} e^{1/x}$

$\lim_{x\to 0^-} e^{\frac{1}{x}} = e^{-\infty} = 0$

$\ln\infty = \infty$

$e^{\infty} = \infty$

$e^{-\infty} = 0$

$\ln(0^+) = -\infty$

- **Horizontal Asymptotes** H.A

The line $y = L$ is called a **horizontal asymptote** of the curve $y = f(x)$ if either

$$\lim_{x\to\infty} f(x) = L \qquad \text{or} \qquad \lim_{x\to-\infty} f(x) = L$$

If $\lim_{x\to\infty} f(x) = L_1$ and $\lim_{x\to-\infty} f(x) = L_2 \Rightarrow y = L_1$ & $y = L_2$ are H.A

Example 2.49. Find the horizontal asymptote of the curve $y = \dfrac{x^2 - 2}{2x^2 + x}$.

① $\lim_{x\to\infty} \dfrac{x^2-2}{2x^2+x} \left(\frac{\infty}{\infty}\right) = \lim_{x\to\infty} \dfrac{x^2\left(1-\frac{2}{x^2}\right)}{x^2\left(2+\frac{1}{x}\right)} = \lim_{x\to\infty} \dfrac{1-\frac{2}{x^2}}{2+\frac{1}{x}} = \dfrac{1}{2}$

② $\lim_{x\to-\infty} \dfrac{x^2-2}{2x^2+x} \left(\frac{\infty}{\infty}\right) = \lim_{x\to-\infty} \dfrac{x^2\left(1-\frac{2}{x^2}\right)}{x^2\left(2+\frac{1}{x}\right)} = \lim_{x\to-\infty} \dfrac{1-\frac{2}{x^2}}{2+\frac{1}{x}} = \dfrac{1}{2}$

$\therefore y = \frac{1}{2}$ is H.A of the curve

Example 2.50. Find the horizontal asymptotes of the curve $y = \tan^{-1} x$.

Example 2.51. Find the horizontal and vertical asymptotes of the graph of the function $f(x) = \dfrac{\sqrt{2x^2+1}}{3x-5}$.

H · A

① $\lim_{x\to\infty} \dfrac{\sqrt{2x^2+1}}{3x-5} = \lim_{x\to\infty} \dfrac{\sqrt{x^2\left(2+\frac{1}{x^2}\right)}}{x\left(3-\frac{5}{x}\right)} = \lim_{x\to\infty} \dfrac{x\sqrt{2+\frac{1}{x^2}}}{x\left(3-\frac{5}{x}\right)}$

$\lim_{x\to\infty} \dfrac{\sqrt{2+\frac{1}{x^2}}}{3-\frac{5}{x}} = \dfrac{\sqrt{2}}{3}$

② $\lim_{x\to-\infty} \dfrac{\sqrt{2x^2+1}}{3x-5} = \lim_{x\to-\infty} \dfrac{\sqrt{x^2\left(2+\frac{1}{x^2}\right)}}{x\left(3-\frac{5}{x}\right)} = \lim_{x\to-\infty} \dfrac{-x\sqrt{2+\frac{1}{x^2}}}{x\left(3-\frac{5}{x}\right)}$

$= \lim_{x\to-\infty} \dfrac{-\sqrt{2+\frac{1}{x^2}}}{3-\frac{5}{x}} = \dfrac{-\sqrt{2}}{3}$

$\therefore$ $y = \pm\dfrac{\sqrt{2}}{3}$ are H · A

V · A

When $3x-5=0 \iff 3x=5 \iff \boxed{x=\frac{5}{3}}$

$\therefore x = \frac{5}{3}$ is V. A

Exercises Page 137 3, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39 (Hint: Squeeze Theorem), 41, 47, 49, 51.

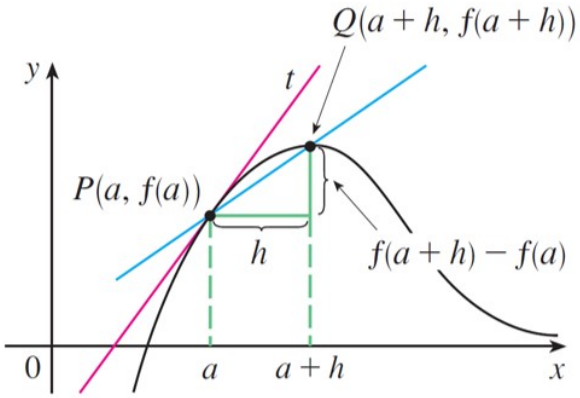
2.7 Derivatives and Rates of Change

- **Tangent Line**

The tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$ is that line through P with slope

$$m_{\tan} = \lim_{h \to 0} m_{\sec} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

provided that this limit exits and is not ∞ or $-\infty$.



Example 2.52. Find an equation of the tangent line to the curve $y = \dfrac{3}{x}$ at the point $(3, 1)$.

- **Instantaneous Velocity**

If an object moves along a coordinate line with position function $f(t)$, then its instantaneous velocity at time $t = a$ is

$$v = \lim_{h \to 0} v_{\text{avg}} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

provided that the limit exists and is not ∞ or $-\infty$.

Example 2.53. Suppose that a ball is dropped from the upper observation deck of the CN Tower, 450 m above the ground.

(a) What is the velocity of the ball after 5 seconds?

(b) How fast is the ball travelling when it hits the ground?

- **Derivative**

The derivative of a function f at a number a is

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

- If the limit exists, we say that f is **differentiable** at a. Finding a derivative is called **differentiation**.

Example 2.54. Given $f(x) = x^3 + 7x$, find $f'(a)$.

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{x \to a} \frac{x^3 + 7x - (a^3 + 7a)}{x - a} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x \to a} \frac{x^3 + 7x - a^3 - 7a}{x - a} = \lim_{x \to a} \frac{(x^3 - a^3) + (7x - 7a)}{x - a}$$

$$= \lim_{x \to a} \frac{(x - a)(x^2 + ax + a^2) + 7(x - a)}{x - a}$$

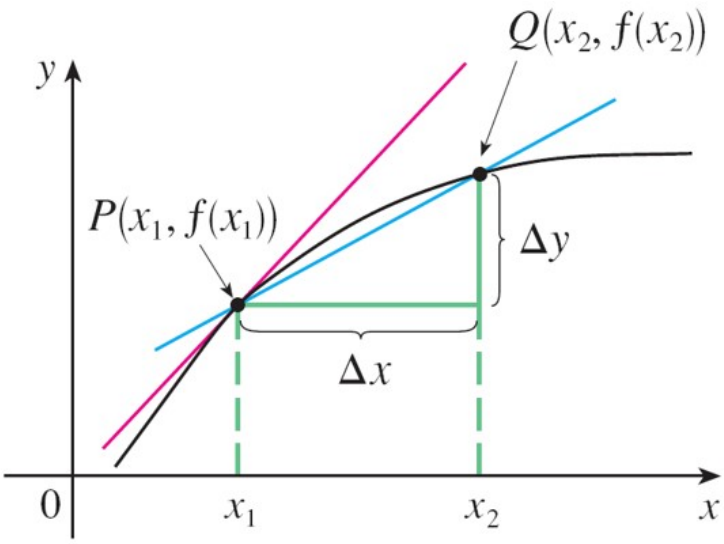
$$= \lim_{x \to a} \frac{(x - a)\left[x^2 + ax + a^2 + 7\right]}{x - a} = \lim_{x \to a} x^2 + ax + a^2 + 7$$

$$= 3a^2 + 7$$

- **Rates of Change**

The instantaneous rate of change of $y = f(x)$ with respect to x at $x = x_1$ is

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{x_2 \to x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \lim_{\Delta x \to 0} \frac{f(x_1 + \Delta x) - f(x_1)}{\Delta x}$$



Exercises Page 148 5, 7, 14 (6.28 m/s, $(10 - 3.72a)$ m/s, 5.4 s, -10 m/s), 21, 31, 33, 35, 37, 39, 41, 55, 59, 60 (exist, why?).

2.8 The Derivative as a Function

- **Definition**

> The derivative of a function f is another function f' whose value at any number x is
>
> $$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

T

Example 2.55. Given $f(x) = \frac{1}{x}$, find $f'(x)$.

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h}$$

$$= \lim_{h\to 0} \frac{\frac{x - x - h}{x(x+h)}}{h} = \lim_{h\to 0} \frac{\frac{-h}{x(x+h)}}{h}$$

$$= \lim_{h\to 0} \frac{-h}{hx(x+h)} = \lim_{h\to 0} \frac{-1}{x(x+h)} = \frac{-1}{x^2}$$

DIY

Example 2.56. Given $f(x) = \sqrt{x}$, find $f'(x)$.

DIY

Example 2.57. Given $f(x) = \dfrac{1}{\sqrt{x}}$, find $f'(x)$. $= -\frac{1}{2} x^{-\frac{3}{2}}$

- **Leibniz Notation for the Derivative**

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = f'(x)$$

- **Other Notations** For $y = f(x)$

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x) = Df(x) = D_x f(x)$$

- A function is differentiable on an open interval if it is differentiable at every number in that interval.

Example 2.58. Where is the function $f(x) = |x|$ differentiable?

① When $x > 0$

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{|x+h| - |x|}{h}$$

We can choose small enough h such that $x+h>0$

$$f'(x) = \lim_{h\to 0} \frac{x+h-x}{h} = \lim_{h\to 0} \frac{h}{h} = 1 \quad \text{exist}$$

$\therefore f(x)$ is differentiable on $(0, \infty)$

② When $x < 0$

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{|x+h| - |x|}{h} =$$

We can choose h small enough such that

$x+h < 0$

$$f'(x) = \lim_{h\to 0} \frac{-(x+h) + x}{h} = \lim_{h\to 0} \frac{-x-h+x}{h} = \lim_{h\to 0} \frac{-h}{h} = -1$$

exist

$\therefore f(x)$ is differentiable on $(-\infty, 0)$

③ When $x = 0$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{|x+h| - |x|}{h}$$

$$= \lim_{h \to 0} \frac{|h|}{h}$$

$h > 0$

$$\lim_{h \to 0^+} \frac{|h|}{h} = \lim_{h \to 0^+} \frac{h}{h} = 1$$

$h < 0$

$$\lim_{h \to 0^-} \frac{|h|}{h} = \lim_{h \to 0^-} \frac{-h}{h} = -1$$

$\neq$

$\therefore \lim_{h \to 0} \frac{|h|}{h}$ D.N.E

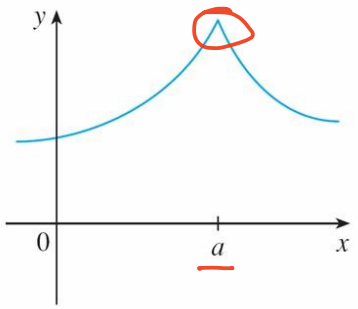
$\therefore f(x)$ isn't differentiable at $x = 0$

- **Differentiability Implies Continuity**

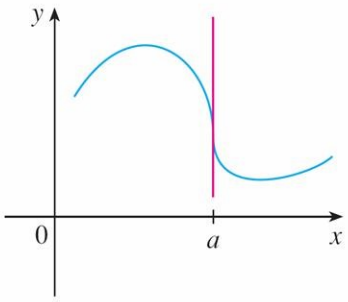
> If $f'(a)$ exists, then f is continuous at a.

- Continuity does **not** imply differentiability.
- Discontinuity implies does not differentiability
- How can a function fail to be differentiable?

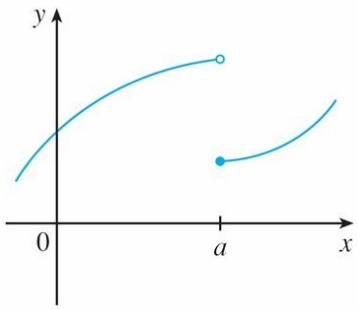
  - A corner



  - A vertical tangent



  - A discontinuity



When the function has a vertical tangent? If satisfied

① $f(x)$ is continuous at $x = a$

② $\lim_{x \to a} f'(x) = \mp\infty$

- **High Order Derivatives**

  The derivative of f is f'. Differentiating f' will produce f'', called the second derivative of f. In the same way, we define f''', the third derivative of f, $f^{(4)}$, the fourth derivative of f, $f^{(5)}$, the fifth derivative of f, and so on.

$[f(x)]^4 = f^4(x) \longrightarrow$ power & $f^{(4)}(x) \longrightarrow$ fourth derivative

Example 2.59. Find all the derivatives of $f(x) = 2x^2 - 6x + 8$.

$f'(x) = 4x - 6$

$f''(x) = 4$

$f'''(x) = 0$

- **Notations**

Derivative	f' Notation	y' Notation	D Notation	Leibniz Notation
First	$f'(x)$	y'	$D_x y$	$\frac{dy}{dx}$
Second	$f''(x)$	y''	$D_x^2 y$	$\frac{d^2y}{dx^2}$
Third	$f'''(x)$	y'''	$D_x^3 y$	$\frac{d^3y}{dx^3}$
Fourth	$f^{(4)}(x)$	$y^{(4)}$	$D_x^4 y$	$\frac{d^4y}{dx^4}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
nth	$f^{(n)}(x)$	$y^{(n)}$	$D_x^n y$	$\frac{d^ny}{dx^n}$

Exercises Page 160 3, 21, 25, 27, 29, 41, 43, 49, 51, 63.